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Perspectives

Translation of Morphological and Functional Musculoskeletal Imaging†

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Abstract:

In an effort to develop quantitative biomarkers for degenerative joint disease and fill the void that exists for diagnosing, monitoring and assessing the extent of whole joint degeneration, the past decade has been marked by a greatly increased role of noninvasive imaging. This coupled with recent advances in image processing and deep learning opens new possibilities for promising quantitative techniques. The clinical translation of quantitative imaging was previously hampered by tedious non-scalable and subjective image analysis. Osteoarthritis (OA) diagnosis using x-rays can be automated by the use of deep learning models and pilot studies showed feasibility of using similar techniques to reliably segment multiple musculoskeletal tissues and detect and stage the severity of morphological abnormalities in magnetic resonance imaging (MRI). Automation and more advanced feature extraction techniques have applications on larger more heterogeneous samples. Analyses based on voxel based relaxometry have shown local patterns in relaxation time elevations and local correlations with outcome variables. Bone cartilage interactions are also enhanced by the analysis of three-dimensional bone morphology and the potential for the assessment of metabolic activity with simultaneous Positron Emission Tomography (PET)/MR systems. Novel techniques in image processing and deep learning are augmenting imaging to be a source of quantitative and reliable data and new multidimensional analytics allow us to exploit the interactions of data from various sources. In this review, we aim to summarize recent advances in quantitative imaging, the application of image processing and deep learning techniques to study knee and hip OA. This article is protected by copyright. All rights reserved

Key Words: osteoarthritis, imaging, deep learning, $T_{1\rho}/T_2$ voxel based relaxometry, bone remodeling, multidimensional data analysis

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1. Introduction

In this review, we present a summary of recent advances in quantitative imaging, the application of image processing and deep learning techniques to study knee and hip osteoarthritis (OA). We aim to give a prospective on how these techniques can help clinical translation of morphological and functional musculoskeletal imaging. We assume a basic knowledge of the current state of the art image acquisition methodologies.

OA is a multifactorial disease that causes joint degeneration, affects 27 million U.S. adults [1; 2], with symptoms such as stiffness, limited joint function and pain which lead to severe disability and impact the overall quality of life. OA primarily affects weight-bearing joints such as the knee and hip joints, and pain is one of the most important outcome measures in OA. Late stage, symptomatic knee or hip OA is treated by total or partial joint replacement, and while this surgically invasive remedy offers relief, joint replacements often fail after 10-15 years, with shorter life spans in obese individuals [3]. Cartilage loss, meniscus changes, subchondral bone changes, ligament, bone marrow changes and changes in other joint tissues are all implicated in OA. Preventive efforts and interventions targeting early stage OA are essential to prevent these tissue level changes, in order to reduce the number of total joint replacement procedures. Thus, identifying subjects at high risk for disease development at an early stage when tissue degeneration is potentially reversible is essential. Ideally, joint degeneration should be proactively prevented with strategies that target the onset and early stages of OA. If subjects at early stages and at high risk of OA are identified, various therapeutic interventions such as lifestyle changes could be advised. Preventative efforts such as weight reduction [4; 5] and various levels of exercise [6; 7] decrease OA progression and should be rigorously enforced if individual assessment demonstrates a high long-term risk for OA.

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OA does not affect just the geriatric population; knee injuries, specifically anterior cruciate ligament (ACL) injuries are well known risk factors for development of post traumatic OA (PTOA) in younger subjects[8; 9]. At the hip joint, well known as a risk factor for OA, femoroacetabular impingement (FAI) is a morphological abnormality of the hip joint, which causes abnormal joint loading patterns and may cause acetabular cartilage delamination [10]. Increased shear forces within the hip joint, particularly due to the cam-type impingement, may cause enlargement of the cartilage flap and lead to complete detachment from the adjacent cartilage thereby producing loose bodies and full-cartilage thickness defects [11].

This alarming burden of knee and hip OA that affects subjects at different age and activity levels calls for the development of quantitative biomarkers for joint degenerative disease to fill the void that exists for early diagnosis, monitoring and assessing the extent of whole joint degeneration.

In addition to radiographs, magnetic resonance (MR) imaging has evolved over the last two decades to respond to this challenge. While MRI can exploit the complexity of the joint with capacity in imaging soft tissues from morphological and biochemical point of views, substantial challenges in image analysis and quantitative image biomarker extraction still hamper clinical translation of promising quantitative techniques widely used in research setting. Lack in standardization of image interpretation, tedious manual post processing pipelines, including image segmentation and registration, feature hand crafting for morphology and relaxometry analysis are just few of the issues related to high-throughput usage of imaging. However, coupled with advanced quantitative imaging techniques, novel computerized image post processing and more recently machine/deep learning techniques the quantitative characterization of early joint degeneration is now a tangible goal.

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While the body of literature on imaging applied to OA is vast, and many are the review articles that aimed to summarize such effort [12,13,14] a void exists in linking recent technical advancements in image processing and deep learning to relevant musculoskeletal clinical questions. With this manuscript, we aim to fill this void by summarizing cutting edge technologies which have plausible promise in changing clinical practice in the near future.

2. Advances in Radiographic Assessment of Knee OA

OA assessment and diagnosis are most commonly done with radiographs (x-rays) using the Kellgren Lawrence (KL) grading system where 0 is normal, 1 shows doubtful signs of OA with potential abnormality, 2 demonstrates definite osteophytes (mild OA), 3 shows definite joint space narrowing (moderate OA), and 4 is severe joint space narrowing with subchondral sclerosis and bony deformity (severe OA)[15; 16]. Although this is a class-based method, the grading system does represent a continues progression of OA, beginning with abnormalities, osteophytes and ending with deformation of the bone. KL grading is widely used for clinical assessment and diagnosis of OA, usually on a high volume of radiographs, however it is still subject to inter- and intra-user variability, making its automation highly relevant [17].

The Osteoarthritis Initiative (OAI) [18] has a large repository of radiographs. 4,504 bilateral PA fixed flexion knee radiographs were collected (age = 61.2 ± 9.2, BMI = 28.6 ± 4.8, male:female = 1886:2618). Subjects were collected from 6 different time points (baseline, 12 months, 36 months, 48 months, 72 months, 96 months) for both left and right knees resulting in a total of 39,593 images. Each of these cases were graded by skilled radiologists involved in the development of the OAI dataset.

Two recent studies used deep learning in an attempt to perform prediction of KL grades analyzing x-rays images. Both the methods used the publicly available datasets obtaining

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comparable performance (quadratic Kappa coefficient of 0.83). Tiulpln at. el. made use of the knee joint symmetry using state of the art Siamese network for the classification. Norman et. al. used a densely connected neural network (DenseNet) to make OA assessments [19; 20]. **Figure 1A** shows the agreement confusion matrix between DenseNet deep learning model and human radiologists in evaluating KL grades on a test set which was never seen in training or tuning phase of the automatic algorithm development. **Figure 1B** depicts 2 examples of saliency maps showing the importance of each pixel in the model decision making process. Performance of both these studies[21 22] are comparable to human reliability. A recent study aimed to determine the reliability of radiographic assessment of knee osteoarthritis (OA) by non-clinician readers compared to an experienced radiologist [23]. This study showed an intra-reader reliability among radiologist (kappa) ranged from 0.40 to 1.0 for individual radiographic features and 0.72 to 1.0 for Kellgren-Lawrence (K-L) grade. Inter-reader agreement among non-clinicians ranged from kappa of 0.45 to 0.94 for individual features, and 0.66 to 0.97 for K-L grade. Inter-reader reliability between non-clinicians and the radiologist ranged from kappa of 0.56 to 0.85 for KL grade. Based on those results the authors suggested that in settings where costs are constrained, trained non-clinicians may be suitable readers of radiographic knee OA. Based on this, deep learning models with weighted kappa equal to 0.83 could also be potentially suitable readers of radiographic knee OA. X-ray is a cheap and fast image modality compared with MRI or CT, however it may lack in sensitivity and specificity. The usage of deep learning for x-ray analysis trained with highly reliable ground truth built by linking different sources (MRI/CT, surgical notes etc), have the great potential of, not just increase throughput in image inspection, but actually increase accuracy, achieving "super-human performances". Additionally, by leveraging on those first experiences in automatic KL grading, deep learning could be potentially used to automatically identify a vast array of other x-rays

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abnormalities (hip OA, fracture etc.). A better understanding of the potential in transfer/meta learning from different anatomies and different pathologies, together with better solutions in incorporating imaging and non-imaging features, will characterize the deep learning x-ray scene in the following years.

Advances in Magnetic Resonance Imaging

Morphological grading

In the last few years Magnetic Resonance Imaging (MRI), with its' ability to provide a rich array of structural and functional features of musculoskeletal tissues, has shed light on disease etiology, potential treatment pathways and predictors of long-range outcomes in OA. Scoring systems, such as the MRI Osteoarthritis Knee Score (MOAKS) [24; 25] and the Whole-Organ Magnetic Resonance Imaging Score (WORMS) [26] have been used to grade local morphological abnormalities in subjects with OA, and compare lesion severity with other findings such as meniscal defects, the presence of bone marrow lesions, as well as radiographic and clinical scores [27-29]. While several studies have showed the value of those semi quantitative grading in relatively large sample sizes (hundreds of subjects*)* the application of this technique is still today hampered by the time and level of expertise necessary to perform reliable grading, in addition, to a lack of actual gold standard and the intra- and inter-observer variability. Previous studies reported radiologist's grading of MRI had a sensitivity of 91.4 % and specificity of 81.1 % in identifying medial meniscal tears, and had 76 % sensitivity and 93.3 % specificity in identifying lateral meniscal tears [30]; sensitivity and specificity were reported to be 74.6% and 97.8% in identifying cartilage lesions [31]. Intra- and inter-observer agreement was 0.87 (0.804 – 0.932) and 0.84 (0.771 – 0.911) for meniscus WORMS and 0.84 (0.771 – 0.911) and 0.79 (0.72 – 0.868) for cartilage WORMS [4].

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Automation of morphological grading of the tissues in the joint would be a significant breakthrough in both OA research and for translation to clinical practice. It would enable the analysis of large patient cohorts and assist the radiologist/clinician in the grading of images. It would change clinical practice with routine incorporation of semi-quantitative grades in radiology clinical reports. This would be a major shift in the paradigm of clinical radiology which could potentially lower the cost in terms of radiologist's time, standardize radiology grading, and ultimately improve patient outcome. Today, although standardized reporting is being initiated in radiology, the terminology is often vague and depends heavily on the radiology practice. Standardization of radiology reports by the usage of structured grading systems in clinical practice may help in a better, more quantitative assessment of disease status and progression. Quantitative data on a larger scale can help better identify patterns and trajectories of specific subgroups and compare similar cases, and eliminates the variability that is introduced by terms as "significant lesion", a descriptor that may have different meaning not only between different practices, but also at different time points in the same practice, or the same radiologist. Identifying patterns in patients' is extremely valuable, for phenotyping; while these patterns may potentially be available today in existing images, they are obscured by the unstructured information hidden in subjective radiology reports, thus making data-mining very difficult.

Using 1487 MRI datasets, a recent study proposed the use of 3D convolutional neural networks to detect and stage severity of meniscus and patellofemoral cartilage lesions in OA and ACL subjects [32]. In this study, a "weakly supervised" method was used by annotating the presence or absence of lesions in the meniscus or patellar cartilage, obviating the need for time-consuming identification of the lesion at the level of image pixel. **Figure 1C** shows the cascade of deep learning networks used in this study to detect image regions of interest and classify the

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presence of abnormalities in cartilage and meniscus. Results from this study showed that the sensitivity was 89.81% and specificity was 81.98% for meniscus lesion detection and sensitivity was 80.0% and specificity was 80.27% for cartilage. **Figure 1D** shows ROC analysis for binary classification. The best performance for staging lesion severity was obtained by including demographic factors, achieving accuracies of 80.74%, 78.02% and 75.00% for normal, small and complex large lesions, respectively. Another study aimed to classify cartilage lesions using similar MRI knee data and deep learning [33]. This method was tested on a smaller dataset of 175 subjects with a 2D patch-based approach and "hard supervision", with two radiologists (readers) identifying of the presence or absence of cartilage lesions such for 2D image patches (64x64) localized to the cartilage region. In this study, the accuracy in binary lesion detection was comparable to [32] , sensitivity and specificity equal to 84.1% and 85.2% respectively for evaluation from the first reader and 80.5% and 87.9% respectively for evaluation from the second reader.

Cartilage Morphology – Volume, thickness:

Cartilage morphology as volume and thickness are often used as OA imaging biomarkers [34]. Loss in medial femorotibial cartilage thickness over 24 moths was shown to be associated with the combination of radiographic and pain progression in the knee; with stronger association for radiographic progression[34]. The processing pipeline of extracting quantitative morphological measurements from MRI often includes a step of image segmentation[35]. An efficient and repeatable fully automatic algorithm for segmentation could establish a standardized practice for identifying imaging biomarkers, as well as the ability to better analyze large datasets in a reasonable amount of time. Manual segmentation is the main obstacle for the translation of cartilage volume and thickness measures to clinical practice. Musculoskeletal tissue automatic segmentation was an unmet challenge for the last several years. However, recent atlas based

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methods and advances in deep learning have brought new end-to-end models that have showed performances never achieved before[35].

Atlas-based algorithms shift the post-processing focus from segmentation to registration. Image registration is the process of defining the best transformation to align two or more images. In atlas-based approaches, a reference is defined by a specific instance in the dataset for which manual segmentation is available or by merging *N* images previously aligned in a common coordinate system. Image registration applies the reference segmentation to the coordinate system of each case that needs to be segmented, typically applying an affine or non-rigid registration. Shan et al. [36; 37] proposed a multi-atlas strategy for both bone and cartilage segmentations, combined with a three-label approach that made use of the atlas segmentations as prior terms in a probabilistic approach. Carbadillo-Gamio et al. [38] also presented an atlas approach and used it for the analysis of thickness and T_2 relaxation times in patella cartilage. This method was then improved and expanded on the overall knee joint and used for the application of voxel based relaxometry [39].

Deep learning based approaches that have produced good results recently are the "deconvolutional", or "encoder-decoder" models [40; 41]. These methods feature a symmetrical network that first learns an encoding by down sampling with convolutions and then learns to decode into a segmentation mask by up sampling with "deconvolutions". Liu et.al. [42] combined Seg-Net, a popular encoder-decoder, with deformable models to perform bone and cartilage segmentation obtaining performance superior to state-of-the-art image processing-based methods. Norman et. al. [43] also used an encoder-decoder, U-net, that features connections between encoding and decoding path allowing for a better combination of local and global features. In this study, different cartilage compartments and meniscus were segmented showing human level

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accuracy and repeatability in extracting cartilage morphology as cartilage and thickness. The automatic procedure proposed in this study produced high resolution 3D segmentation of cartilage and meniscus ~5 seconds. Average testing correlations between manual and automatic quantification of $T_{1\rho}$ and T_2 relaxation time values were 0.8233 and 0.8603, respectively, and 0.9349 and 0.9384 for volume and thickness, respectively. Longitudinal precision of the automatic method was comparable with the manual one. **Figure 1E** shows a comparison between manual and automatic segmentation

Even though MRI is considered a sensitive and specific tool for the identification of OA degenerative changes, we still lack an actual gold standard. The uncertainty in image annotations/grading is a point of extensive discussion in the field of deep learning applied to medical imaging [44]. Recent literature reports examples of applications of Bayesian drop out techniques to model uncertainty applied to neuroimaging [45] and experiments showing the robustness of deep learning to label noise [46]. In most of the medical imaging applications there is a lack of a real gold standard and before effective clinical translation, there is a need for data harmonization and the definition of a common ground truth between panels of experts.

Robustness and generalization of these techniques when applied on images acquired using different MRI sequences or those acquired on different (vendor) systems still needs to be tested. Mechanisms of continuous learning will need to be designed in the initial phases of clinical usage. This however, raises several questions, for example, the levels of reliability of the new ground truth coming from the daily use would depend on individual user radiologist preferences and training, and methods to handle Artificial Intelligence techniques with this sort of reinforcement learning will need to have a defined evolution methodology to avoid instability.

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3. Quantitative MRI To Extract Biochemical Compositional Features

While the analysis of MRI derived morphological OA grades coupled with novel image processing and deep learning techniques have great potential, it is well known that morphological changes are preceded by changes in the cartilage extra-cellular matrix, which are not captured by the metrics above. Quantitative MR, including $T_{1\rho}$ and T_2 mapping has been extensively used to probe biochemical changes in the articular cartilage in the early stage. While T_2 relaxation time is primarily affected by hydration and collagen structure due to dipolar interactions [47], the spin-lock techniques used in $T_{1\rho}$ reduce dipolar interactions [48]. Chemical exchange on proteoglycan and water protons was suggested to contribute to $T_{1\rho}$ in cartilage, although $T_{1\rho}$ changes in cartilage may be affected by hydration and collagen structure as well [49]. The task of determining quantitative MRI degenerative changes is usually accomplished through region of interest (ROI)-based approaches [50-54] . In this class of techniques, compartments of the cartilage are segmented and each ROI is described by average encompassed relaxation time values. Sensitivity of relaxation time evaluation trough the extraction of simple averages of cartilage global compartments was widely discussed and previous studies reported that spatially assessing MR images of the knee cartilage relaxation times using laminar and sub-compartmental analyses could lead to better and probably earlier identification of cartilage matrix abnormalities [55; 56].

While the value of quantitative MRI techniques is extensively proven by the body of recent literature [57-59], the clinical translation of these advanced imagining techniques is still hampered by the tedious and non-scalable post processing or image segmentation required and the too simplistic feature extraction techniques used for the quantification. Accordingly, the last few years of quantitative MRI research were characterized by a growing interest in exploring fully automatic techniques to assess spatial distribution and local patterns in relaxation time maps. Extraction of

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second order statistical information or texture analysis [60; 61] has been widely used to overcome the limitation of the average ROI-based approaches with promising results. However, texture analysis does not address the problem of regional or compartmental differences between the two groups, does not enable the extraction of salient relaxometry patterns and still require segmentation.

A fully automatic, local and unbiased algorithm for studying knee $T_{1\rho}$ and T_2 relaxation times by creating an atlas and using voxel-based relaxometry (VBR) was recently proposed [39]. This technique allows for the investigation of local cartilage compositional differences between two cohorts, or between different time points in the same cohort. An atlas-based technique aligned all images onto a single template and was used for fully automatic articular cartilage segmentation[39]. VBR has been used in a multicenter study with the aim to explore how cartilage lesions at the time of ACL injury influence the longitudinal progression of cartilage degeneration [62], to study differences between the ACL injured and contralateral knee in the composite R_2-$R_{1\rho}$ metric and to explored this metric as a possible associative biomarker for patient-reported outcomes[63]. VBR coupled with principal component analysis (PCA) has been used to compare biochemical patterns in ACL and OA subjects [64]. VBR was also employed in analyzing the effect of posterior meniscus lesions on adjacent cartilage composition, in absence of morphological cartilage defects[65], and correlation between relaxation time and patellofemoral joint stress in PFJOA subjects [66]. **Figure 2A** and **Figure 2B** show two examples of VBR group analysis. In **Figure 2A** group average for 64 ACL subjects on the injured side, the corresponding contralateral side and the paired average % difference is shown; in **Figure 2B** group average for 49 patello-femoral osteoarthritis (PFJOA) subjects, 34 matched controls, average % difference and p-value map is shown. **Figure 2C** shows an example of local correlation analysis with a non-imaging variable. Local R-value and p-value maps depict the associations between T_2 values and PFJOA stress computed from gait kinematic

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data.

The spatial variation of cartilage relaxation times in the hip joint, and the need for a regional analysis to extract imaging biomarkers from relaxation parametric maps has also been recognized [67; 68]. In an attempt to address this challenge, VBR has been also proposed for the hip joint and was adopted to study the local association between $T_{1\rho}$ and T_2 relaxation times and longitudinal changes in patient-reported outcomes (**Figure 2D)** [69]. Combination of texture analysis and VBR was recently proposed as imaging biomarker to detect delamination in subjects with FAI [70].

Automatic VBR analysis makes clinical translation of quantitative compositional MRI a more tangible goal, bringing the reality of characterizing early changes in the extracellular matrix closer to the poInt of patient management. However, challenges related to pulse sequence and hardware sensitivity of relaxation time measures are still open questions that need to be addressed.

Analysis of Bone Remodeling

In addition to degeneration of soft tissues, it has been suggested that changes also occur in the subchondral and trabecular bone. Articular cartilage and subchondral bone act in tandem with regards to the mechanical loading of the joint. The subchondral mineralized zone plays an important role in reducing the impact forces typically encountered during dynamic joint loading and adapts to the mechanical demands during normal and abnormal joint loading [71-73]. Both early-stage increased bone remodeling and bone loss, as well as, the late-stage reduced bone remodeling and subchondral densification are important components of the pathological process that leads to OA.

Bone Shape

Looking at bone remodeling macroscopically, several investigators have proposed that aspects of bone shape are associated with an increased risk of incident OA and with the severity

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of OA, based on anthropometric measures, cross-sectional findings, or shape modeling of knees [74-77]. Studies based on 2D radiographs also reported bone shape gender differences, an association with incidence OA [78] and differences in bone shape in subjects with lateral and medial OA [79]. Statistical Shape Modeling (SSM) is a widely-used tool to describe shapes in a brief but comprehensive features vector. SSM has the ability to characterize complex shapes using Principal Component Analysis (PCA) to reduce the dimensionality of the data. PCA provides an orthonormal basis. Therefore, each component of the features vector (mode) describes a different aspect of the bone shape uncorrelated with the other components. The effect of each mode on the average surface can be modeled individually, synthesizing new instances. SSM has the capability to analyze shape differences without *a priori* assumptions, instead identifying the geometrical features empirically. SSM coupled with the 3-dimensional nature of MRI has great potential in identifying knee OA risk factors and in studying disease pathogenesis and large body of work was done recently [75; 77; 80-82]. Other studies also used this technique to evaluated the contribution of knee shape to ACL tears[83], to assess the association between bone shape and the progression of cartilage degeneration[84] and alternated knee kinematics [85] after ACL reconstruction. **Figure 3A** shows modeling of the shape variation associated with abnormal knee kinematics after ACL reconstruction.

The ability of 3D SSM to quantify hip bone shape features from MRI was also recently published. In this study bone shapes were associated with morphological and biochemical degeneration of the articular cartilage and may be early signs of hip OA. **Figure 3B** shows VBR analysis of the relationship between coxa valga shape variation of the proximal femur and prolongation in $T_{1\rho}$ relaxation time in subjects with hip OA. Associations between bone shapes with hip joint mechanics during gait were also observed. This study confirmed previous results

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that used radiographs to establish a link between proximal femur bone shapes and OA [86; 87], showing further evidence of the critical role of the bone shape in the development of OA and of the ability of MRI and 3D shape modeling to quantify those features.

Cartilage-Bone Connectome

Bone cartilage crosstalk is consider as one of the key factors in understanding OA pathogenesis[88]. A hypothetical model for OA pathogenesis has been proposed by Burr et. al. [89], whereby repetitive joint loading causes an initial increase in bone remodeling, that is associated with increased vascular invasion of the deep layers of cartilage, which allows access to the cartilage by chondrolytic enzymes unopposed by inhibitors of the degradative proteinases, that cause a breakdown of the extra-cellular matrix, loss of proteoglycan and collagen, and thus a loss of cartilage compressive stiffness, and an overload of the joint. Subchondral bone changes are present prior and during development of OA and increased bone blood flow and bone remodeling as demonstrated by ^{18}F Sodium Fluoride (NaF) positron emission tomography (PET)-computed tomography (CT) [90] may be associated with patellofemoral pain and later stage morphological changes in cartilage.

Simultaneous PET-MRI was also recently used to study cartilage and bone interactions [91 92-95]. Savic et. al. [95] performed a pilot study analyzing sixteen OA subjects and showed increased uptake of NaF in the knee in OA subjects. Similar results were then obtained by Kogan at. el., by analyzing 22 subjects with knee pain or injury. Both studies showed increase in Standard Uptake Values (SUV) in proximity of morphological abnormalities as bone marrow edema. **Figure 3D** shows an example of a subject with knee pain and bone marrow edema like lesion (BMEL) abnormality with clear localized increased PET uptake in correspondence to the morphological abnormality. Savic et. al. [95] also explored correlations between voxel based relaxation time $T_{1\rho}$ and

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T_2 in the cartilage with sodium fluoride uptake metrics (slope K_i and SUV), showing correlations not only not only in adjacent regions but also in compartments that are not adjoining. Correlations between SUV and cartilage compositional data were also recently shown in ACL subjects [93].

The ability to simultaneously monitor early changes in bone remodeling and cartilage composition, linked with the assessment of morphological shape changes has the potential to make a tremendous impact of defining OA as a total joint disease. The translation of these tools to the clinic would potentially bring not just advanced diagnosis to the point of patient care, but also open the path for drug discovery and the development of new therapeutic regimes.

4. Multidimensional Data Analysis

Advanced imaging and deep learning techniques discussed above allow us to extract an extraordinary rich array of heterogeneous information on populations with large sample sizes. While the complexity of knee and hip degeneration as a whole joint disease, may be captured the data analytics and tools, more commonly used, are often limited to either univariate analysis or use a subset of the variables. Innovations in the big data analytics field have brought several multidimensional visualization and analytics methods with which we can compare individual patients as a 'point-cloud' in multidimensional space, overcoming the inherent limitations of single endpoints. One possible approach considers the usage of data shape or topology [96]. Topological data analysis (TDA) is a branch of applied mathematics aimed at analyzing relationships between data and data shapes. In essence, this involves projecting individual patients into the 'syndromic space' defined by all outcome variables simultaneously [97]. The fundamental idea of TDA is that this method acts as a geometric approach to pattern recognition within data. By extracting fundamental shapes (patterns) in high-dimensional data TDA provides visualization that can provide novel insights about the data and can identify meaningful sub-groups [96]. Applications of

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TDA to study Knee [98] and Hip [99] OA were recently presented showing the usage of the technique as a descriptive tool able to identify subject sub groups and explain some of the complex interactions between imaging and biomechanics and relationships between cartilage degeneration and symptoms. Considering simultaneous analysis of MRI and biomechanics data the study presented in [98] showed a syndromic network with a clear topology composed by 3 main sub-groups (**Figure 4A**). Significant KL differences were observed between sub-groups: sub-group 1 (left) 0.79 ± 0.95, sub-group 2 (right) 1.29 ± 1.17 p-value = 0.010) (**Figure 4B)**, however, T_2 relaxation time in the anterior lateral femur was observed as the strongest sub-group predictor (p-value = $3.98*10^{-11}$). the ability of the TDA to reveal insights and generate hypothesis regarding which imaging parameters maybe able to predict disease progression was also assessed identifying the composite metric $1/T_2$-$1/T_{1\rho}$ = R_2-$R_{1\rho}$. KOOS sport sub scores and joint effusion measured by WORMS as strongest predictors of cartilage lesions worsening (**Figure 4C**). A similar approach was applied to study trait of hip OA. In this study, significant association between cartilage degeneration and pain was observed in subjects without any sign of OA and subjects with radiographic hip OA (**Figure 4D**). An inverted relationship was observed at the early stage of the disease, suggesting that pain protective skeletal loading alterations may contribute to faster degeneration. Analysis of sub subgroups identified knee biomechanics as an initial marker of the disease, noticeable before the morphological progression and degeneration (**Figure 4E** and **Figure 4F)**. Further analysis of an OA subgroup with femoroacetabular impingement (FAI) showed anterior labral tears to be the most significant marker (P = 0.0017) between those FAI subjects with and without OA symptoms.

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5. Conclusion and Future Directions

In conclusion, quantitative imaging is growing as an area of research. The advent of machine/deep learning methods may potentially be used for standardized grading of x-rays, MR morphological changes and disease progression. The standardization of quantitative imaging, relaxation time imaging lays the fundamental groundwork for characterizing early changes in cartilage, meniscus, ligaments, with the inclusion of PET methods, further expands the realms to include bone remodeling. These methods when combined with multi-dimensional, big-data analytics further opens up the horizons for understanding the etiology and progression of joint degeneration and osteoarthritis.

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Figures Legends

Figure 1: Artificial Intelligence to extract OA features from X-ray and MRI: A. Agreement Confusion matrix between a deep learning model and human radiologists in evaluating KL grade on a test set of x-ray images from OAI. **B.** Saliency Maps computed as gradient of output category with respect to input image; Saliency Maps show how output category value changes with respect to a small change in input image pixels since they show the importance of each pixel in the model decision making process. In the first raw the hot spot is in correspondence to the location of and osteophyte and in the second raw it is in the correspondence of joint space narrowing. **C.** model architecture used for simultaneous multi-tissue segmentation and meniscus and cartilage anomaly detection. **D.** ROC curve showing deep learning model performance in detecting presence of cartilage and meniscus lesions. **F.** comparison between manual and automatic segmentation produced by a U-net deep learning model.

Figure 2: Voxel Based Relaxometry. A. Example of $T_{1\rho}$ Voxel based paired group analysis: group average for 64 ACL subjects in injured side, corresponding contralateral side and paired average % difference **B.** Example of $T_{1\rho}$ Voxel based unpaired group analysis: group average for 49 PFJOA subjects, 34 matched controls, average % difference and p-value map showing significant elevation in PFJOA subjects. **B.** Example of Voxel based analysis of the associations between local T_2 and PFJOA stress showing significant associations in overall population (first raw) which is driven by PFJOA population (second row), controls are shown in the third row.

D. Example of VBR-hip: rho-value and p-value maps showing significant correlation between acetabulum $T_{1\rho}$ and ΔHOOS over 18 months.

Figure 3: Analysis of Bone Remodeling. A. Example of 3D MRI-based Statistical Shape Modeling (SSM) in ACL subjects, showing shape variations associated with abnormal joint

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kinematic after ACL reconstruction **B.** Voxel based analysis of the associations between proximal femur bone shape variations and cartilage T1r relaxation time. Subjects with coxa valga variation showed significant increase relaxation time both femoral and acetabulum. **C.** Analysis of bone remodeling with ^{18}F Sodium Fluoride (NaF) PET/MR. this example shows SUV uptake in OA subject with bone marrow edema and knee pain.

Figure 4: Multidimensional Data Analysis using Topology: *first row shows an example of Topological Data Analysis to study interactions between cartilage degeneration and biomechanics data in a dataset of 178 Knee OA patients***. A.** Extracted network based on biomechanics and compositional MRI variables showing the presence of three distinct sub-networks marked with dashed circles. **B.** The same network is shown colored by Kellgren and Lawrence (KL) grade (OA severity). The combined network showed differences in osteoarthritis severity between the subnetwork 1 (prevalence of blue nodes low KL grading) and subnetwork 2 (prevalence of red nodes high KL grading). **C.** the subjects in the progression cohort are located in both subnetworks. However, In subnetwork 1 progressors are located in a specific region marked with a dashed circle. Combined metric R2-R1r is the first variable able to significantly distinguish between progressors within subnetwork 1 and the rest of subnetwork 1.

Second row shows an example of Topological Data Analysis to study associations between progression in cartilage degeneration and symptom worsening 128 Hip OA patients **D.** Extracted network based on biomechanics, bone shape and compositional MRI variables showing colored by Kellgren and Lawrence (KL) grade (OA severity). The combined network showed differences in osteoarthritis severity in three subnetworks.

E. Topology colored by percentage of subjects for each node that showed significant T1r/T2 progression in 36 month "compositional progression". **F.** Topology colored by percentage of

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subjects for each node that showed significant HOOS progression in 36 month "symptoms progression". Dashed circle showed opposite trends between compositional and symptoms progression in the OA early stage subnetwork.

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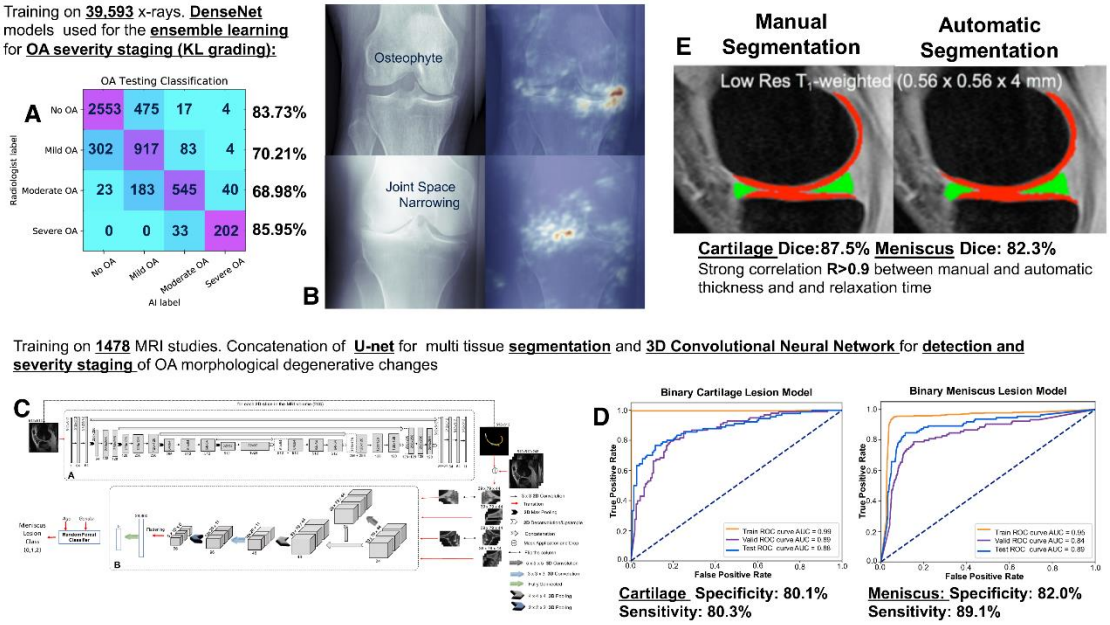


Figure 1

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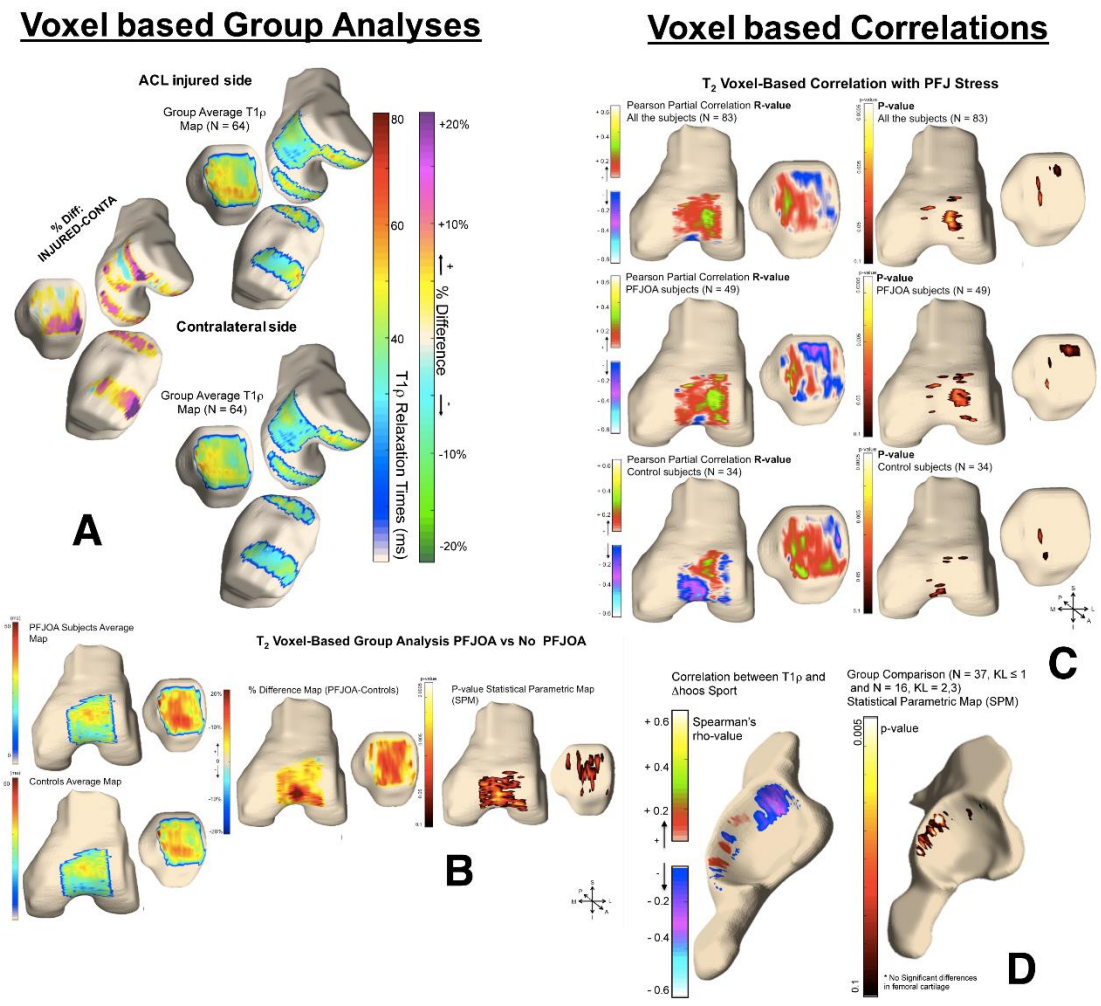


Figure 2

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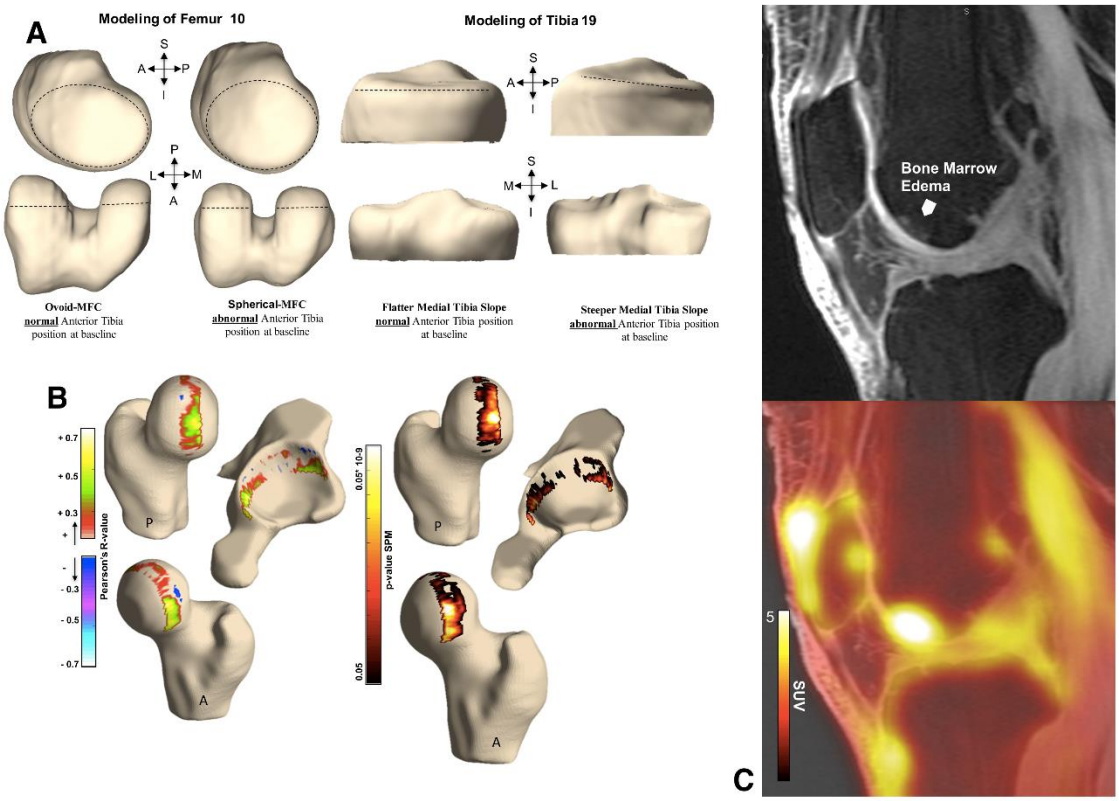


Figure 3

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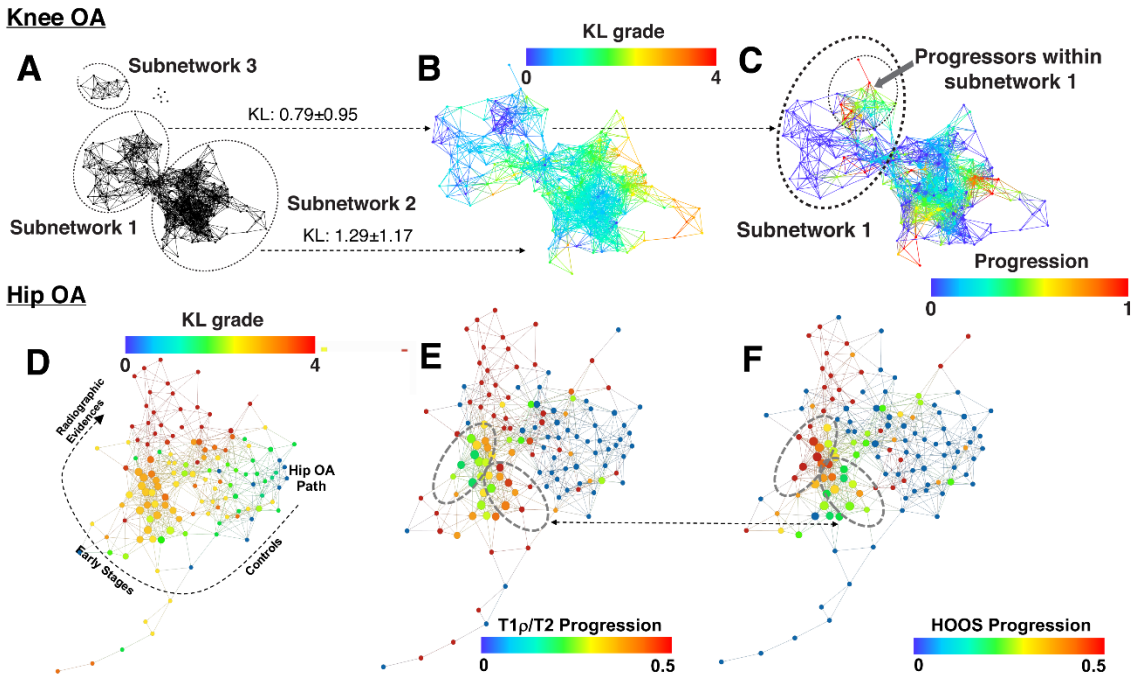


Figure 4

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